

Customer Churn Analysis Report

OTT Subscription Platform — SQL & Python Analytics Project

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Tools used:	Python (pandas, numpy), matplotlib, seaborn, SQL
Dataset:	21 customer records — subscription, billing, and support attributes
Analysis type:	Descriptive + diagnostic churn analysis with risk segmentation

Executive Summary: Of 21 subscribers analyzed, 6 churned — a 28.6% churn rate. The single strongest driver identified is contract length: monthly-contract customers churn at 55.6%, nearly seven times the 8.3% rate seen in annual-contract customers. Basic-plan subscribers also churn disproportionately (60%). Customers who escalated a support issue churned at 100% in this sample. These patterns point to a clear, actionable retention lever: migrating monthly subscribers to annual plans and treating support escalations as an early-warning signal.

1. Business Objective

Customer retention is the primary growth lever for subscription-based platforms — acquiring a new subscriber costs significantly more than retaining an existing one. This analysis examines an OTT subscriber base to answer three questions: **who** is churning, **why** they churn, and **where** the business should focus retention effort for the greatest return.

2. Dataset & Methodology

The dataset contains 21 subscriber records spanning subscription details (plan type, contract type, billing, renewal/cancellation dates), demographics (state, gender), and customer support interactions (complaints, escalations, satisfaction scores). Support-interaction data was recorded only for customers who filed at least one complaint (7 of 21 customers) — this is noted wherever it affects interpretation.

Workflow:

- Loaded and inspected the dataset for structure, data types, and missing values
- Cleaned data — parsed date fields, standardized categorical fields, handled nulls in support-related columns
- Engineered features — churn flag, customer tenure (days), a 3-tier churn risk category (Low/Medium/High) from churn score
- Ran group-by aggregation across plan type, contract type, state, gender, and support-escalation status
- Computed core retention KPIs — churn rate, ARPU, revenue at risk, CLTV lost, escalation correlation
- Visualized results with matplotlib/seaborn to surface the clearest patterns

3. Key Metrics

Metric	Value
Total Customers	21
Churned Customers	6
Churn Rate	28.57%
Retention Rate	71.43%
ARPU (Average Revenue per User)	Rs 18.85 / month
Monthly Revenue at Risk (from churned accounts)	Rs 73.94
CLTV Lost to Churn	Rs 2,047 (of Rs 17,294 total CLTV)
Average Tenure — Churned Customers	773 days
Average Tenure — Active Customers	1,223 days

4. Visual Analysis

4.1 Churn by Contract Type — the strongest signal

Monthly-contract subscribers churn at **55.6%**, compared to just **8.3%** for annual-contract subscribers — a **6.7x** gap. This is the single clearest, most actionable pattern in the dataset.

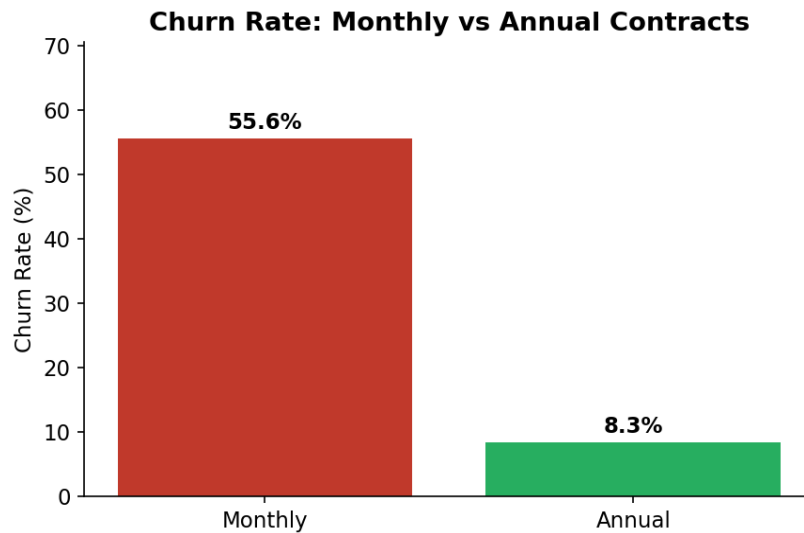


Figure 1: Churn rate by contract type

4.2 Churn by Plan Type

Basic-plan subscribers churn at **60%**, far above Standard (22.2%) and Premium (14.3%). Since Basic is the lowest-priced tier, this churn is high in rate but comparatively lower in revenue impact per customer.

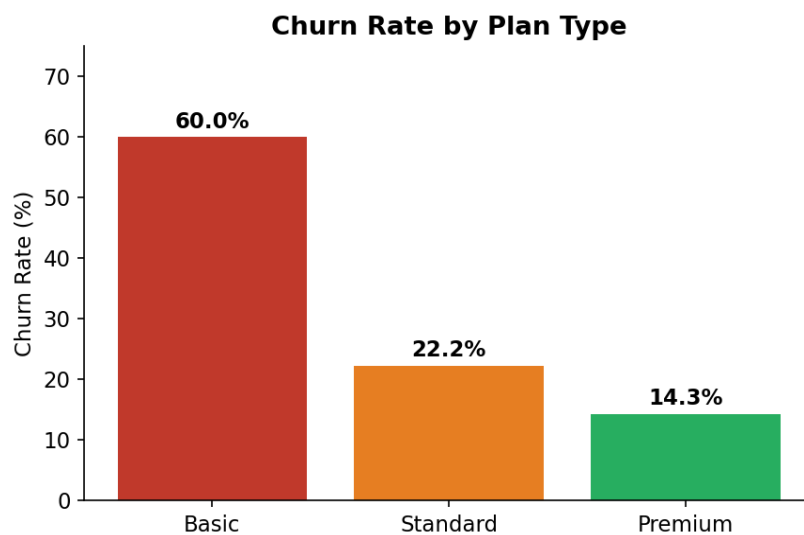


Figure 2: Churn rate by subscription plan

4.3 Churn Risk Score Validation

The churn_score field (used to segment customers into Low/Medium/High risk) tracks perfectly with actual outcomes in this sample: every customer scored 'High' risk churned, while every 'Low' and 'Medium' risk customer stayed. This validates using churn_score as the primary prioritization signal for retention outreach.

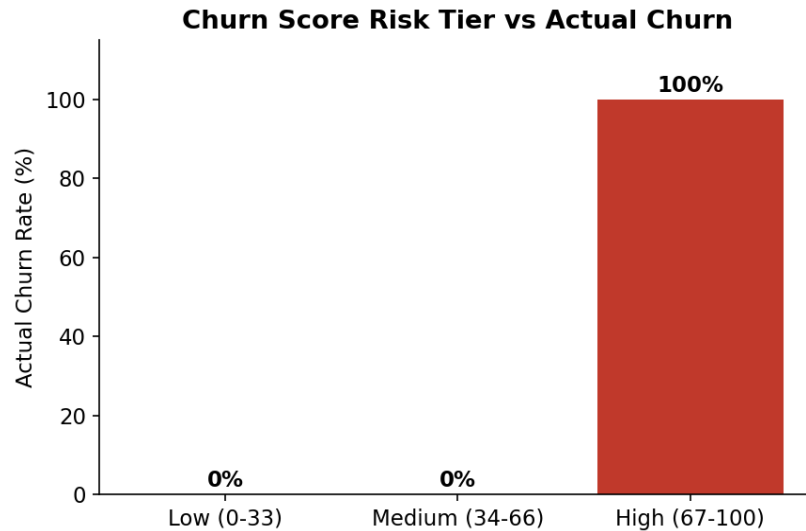


Figure 3: Churn score risk tier vs. actual churn outcome

4.4 Churn by State

Karnataka shows the highest churn concentration in the sample. Note the small sample size per state (shown as 'n' below each bar) — this should be treated as a lead to investigate, not a confirmed regional trend.

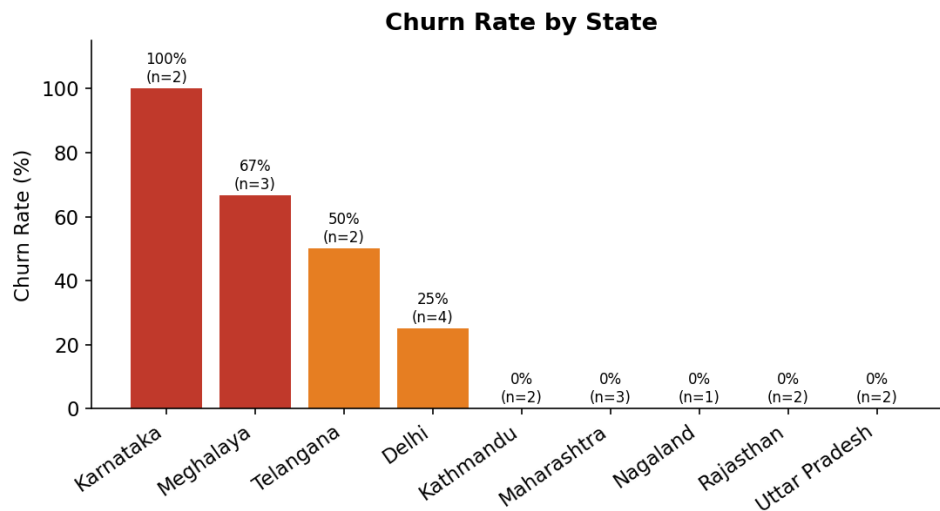


Figure 4: Churn rate by state (n = customers per state)

4.5 Churn by Gender

Female subscribers churned at **33.3%** versus **22.2%** for male subscribers. The gap exists but the sample per group is small, so this is worth monitoring rather than treating as conclusive.

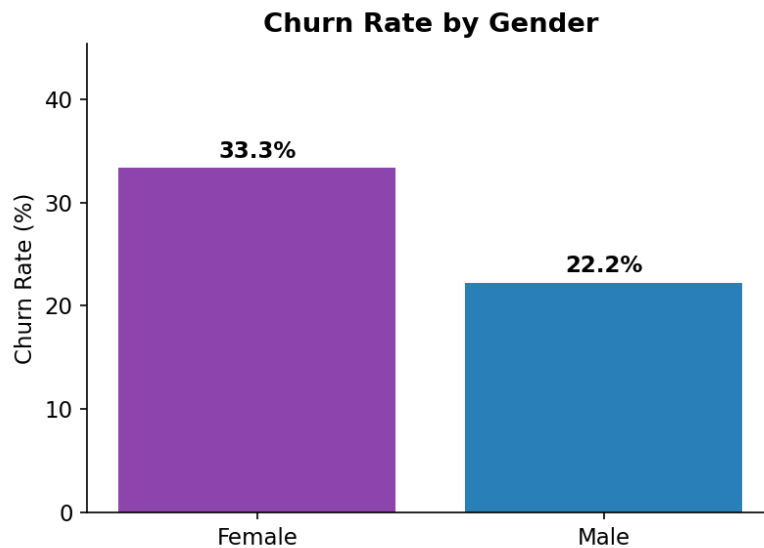


Figure 5: Churn rate by gender

4.6 Revenue Split — Retained vs At Risk

Of Rs 395.79 in total monthly recurring revenue across all 21 customers, Rs 73.94 — about 19% — sits with already-churned accounts and is no longer being collected.

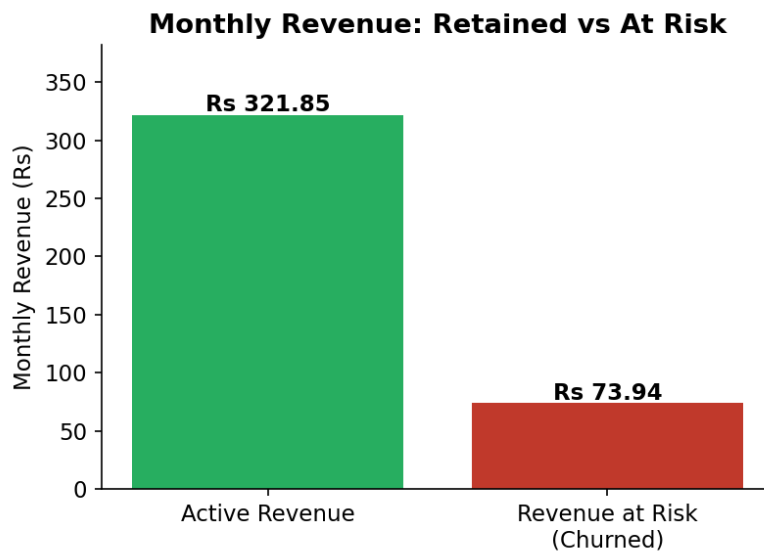


Figure 6: Monthly revenue — retained vs. churned accounts

4.7 Why Customers Cancel

Among customers who gave a cancellation reason, **switching to a competitor** was the most common driver, followed by price sensitivity and content/quality dissatisfaction in roughly equal measure.

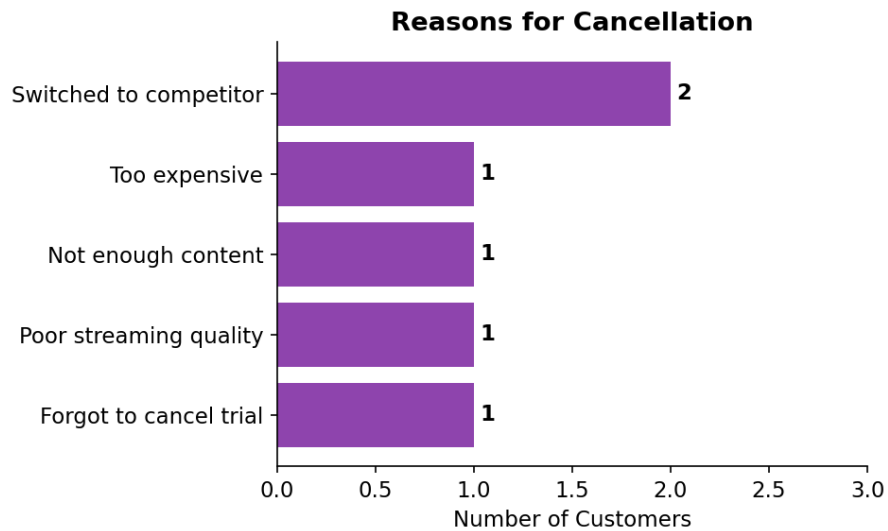


Figure 7: Stated reasons for cancellation

4.8 Cancellation Timing

Cancellations were not evenly spread across the year — **September 2024** saw a spike (2 of 6 total cancellations), suggesting a triggering event that month rather than steady background attrition.

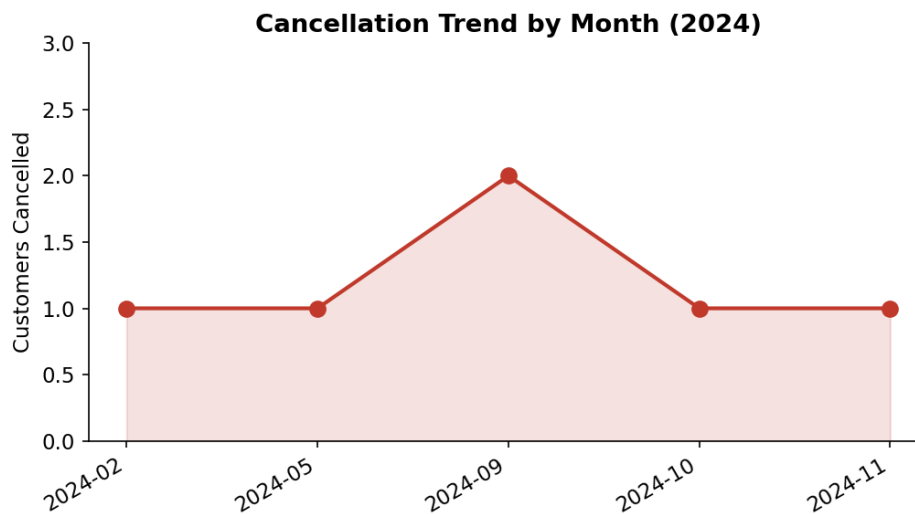


Figure 8: Monthly cancellation trend, 2024

5. Support Interactions as a Churn Signal

Support data (complaints, escalations, CSAT) exists only for the 7 customers who filed at least one complaint — 6 of these 7 churned. Within this subset, customers whose complaint was escalated churned at **100%** (4 of 4), versus **67%** (2 of 3) for non-escalated complaints.

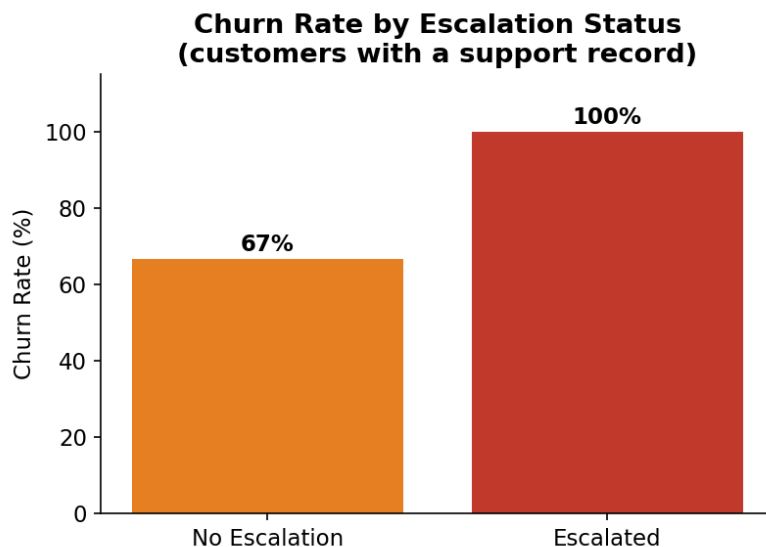


Figure 9: Churn rate by escalation status

Average CSAT among churned customers with support records was **34/100**, notably lower than the single active customer with a support record (90/100).

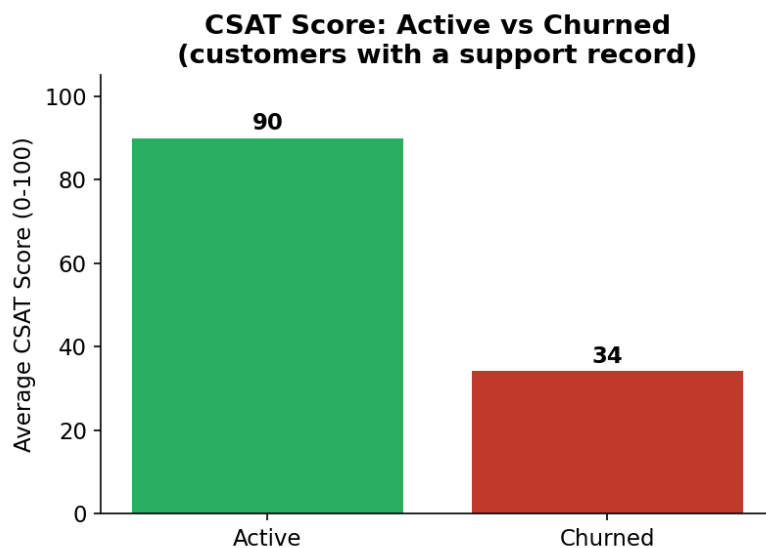


Figure 10: CSAT score — active vs. churned (customers with a support record)

Caveat: this subset is small (7 customers), so treat this as a directional signal worth monitoring at scale rather than a statistically confirmed relationship.

5.1 Tenure — Active vs Churned

Churned customers had a shorter average relationship with the platform (773 days) compared to active customers (1,223 days) — churn risk appears elevated for newer customers, not just at random.

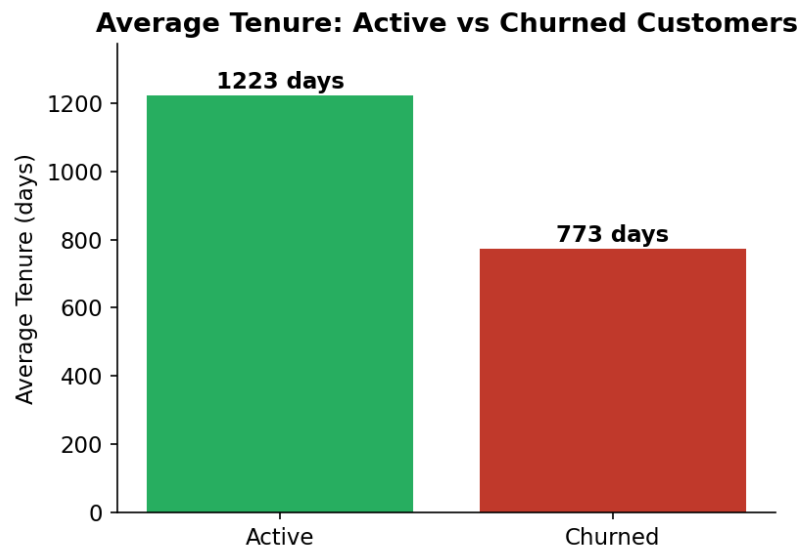


Figure 11: Average customer tenure — active vs. churned

6. Key Insights

- Contract type is the strongest churn lever in the data: monthly subscribers churn 6.7x more than annual subscribers.
- The churn_score risk model perfectly separates churned from retained customers in this sample — a strong candidate for prioritizing retention outreach.
- Support escalation is an early-warning signal: 100% of escalated-complaint customers churned, versus 67% of non-escalated ones.
- Basic-plan churn is high in rate (60%) but limited in revenue impact given its low price point relative to Premium.
- Cancellations cluster in September 2024 and cite competitor switching most often — pointing to a possible competitive or pricing event that period.
- Churned customers had notably shorter average tenure (773 days) than active customers (1,223 days), suggesting churn risk is not purely random but tied to a shorter relationship with the platform.

7. Business Recommendations

- Launch a monthly-to-annual migration campaign (e.g. discount incentive) — the highest-leverage lever identified, given the 6.7x churn gap.
- Use churn_score to build a proactive retention queue, prioritized by CLTV, rather than waiting for cancellation requests.

- Route any escalated support ticket into a retention workflow immediately — treat escalation as a leading indicator, not just a service issue.
- Investigate the September 2024 cancellation spike specifically — check for a price change, service disruption, or competitor promotion in that window.
- Monitor CSAT proactively rather than only after a complaint — the sharp CSAT gap between churned and retained customers suggests satisfaction erodes before cancellation.

8. Tools & Techniques

Data handling: pandas, numpy | **Analysis:** group-by aggregation, correlation analysis, risk-tier segmentation | **Visualization:** matplotlib, seaborn | **Source integration:** SQL (multi-table joins on customer ID)